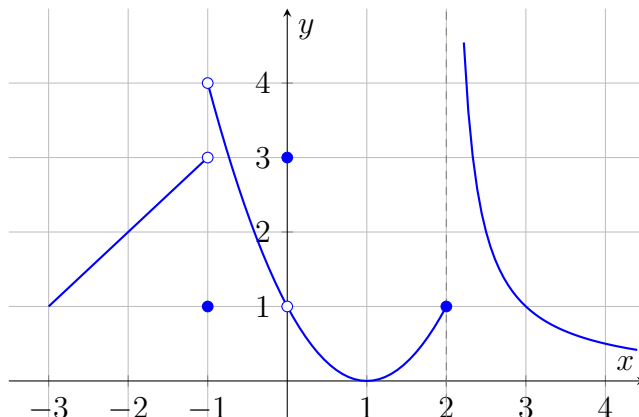


Math 1512 Calculus 1 Review
Fall 2026

1. Formulas to know.

- (a) The definition of “ f is continuous at $x = a$ ”
- (b) $f'(a)$, the derivative of f at $x = a$, written as a limit in two ways (as $x \rightarrow a$ and as $h \rightarrow 0$)
- (c) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$ and $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}$
- (d) Derivatives of $\sin(x)$, $\cos(x)$, $\tan(x)$, $\sec(x)$, $\csc(x)$, $\cot(x)$
- (e) Power rule, product rule, quotient rule, chain rule
- (f) The linearization $L(x)$ of f at $x = a$, and the differential dy
- (g) Definition of a critical number of f
- (h) State the Intermediate Value Theorem, the Extreme Value Theorem, Fermat’s Theorem, and the Mean Value Theorem
- (i) First derivative test, second derivative test, and the definition of an inflection point
- (j) Antiderivatives of x^n ($n \neq -1$), $\sin(x)$, $\cos(x)$, $\sec^2(x)$, $\sec(x) \tan(x)$
- (k) The definite integral $\int_a^b f(x) dx$ as a limit of Riemann sums
- (l) Both parts of the Fundamental Theorem of Calculus
- (m) The average value of f on $[a, b]$
- (n) Volume of a cylinder, a cone, and a sphere. Area of a circle. Surface area of a sphere.

2. Limits from a graph. Let g be the function graphed below. A filled dot is a point on the graph, an open dot is not.



- $g(-1)$, $\lim_{x \rightarrow -1^-} g(x)$, $\lim_{x \rightarrow -1^+} g(x)$, $\lim_{x \rightarrow -1} g(x)$
- $g(0)$ and $\lim_{x \rightarrow 0} g(x)$
- $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$
- At which x values in $(-3, 4)$ is g not continuous? For each, say whether the discontinuity is removable, a jump, or infinite.
- At which of these x values is g continuous from the left? From the right?
- Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} h(x) = -2$. Use the limit laws to evaluate $\lim_{x \rightarrow a} [3f(x) - 5h(x)]$, $\lim_{x \rightarrow a} f(x)h(x)$, $\lim_{x \rightarrow a} \frac{f(x)}{h(x)}$, and $\lim_{x \rightarrow a} \left(\sqrt{f(x)} + h(x)^2 \right)$.

3. Computing limits. Evaluate the limit, or explain why it does not exist. Use ∞ or $-\infty$ when appropriate.

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$

(b) $\lim_{x \rightarrow 5} \frac{\sqrt{x+4} - 3}{x - 5}$

(c) $\lim_{x \rightarrow -1} \frac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$

(d) $\lim_{x \rightarrow 4} \frac{5}{x^2 - 3x - 4}$ (consider the one-sided limits)

(e) $\lim_{x \rightarrow -1} \frac{x}{(x+1)^2}$

(f) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$

4. Squeeze Theorem.

(a) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$

(b) If $2x - 1 \leq f(x) \leq x^2 - 2x + 3$ for all x , find $\lim_{x \rightarrow 2} f(x)$.

(c) Explain why the product law cannot be used to evaluate the limit in (a).

5. Limits at infinity and asymptotes.

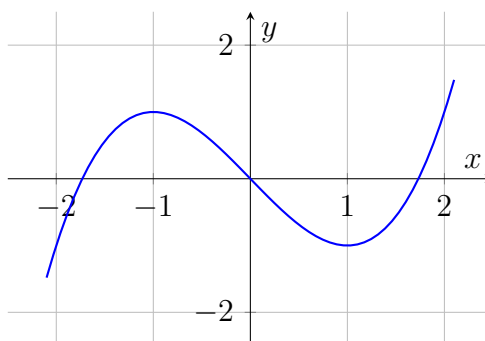
- (a) $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 2}{5x^2 + 4}$
- (b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$ (Recall $\sqrt{x^2} = |x|$)
- (c) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$
- (d) Find all vertical and horizontal asymptotes of $f(x) = \frac{2x^2 - 2}{x^2 + x - 2}$. Why is $x = 1$ not a vertical asymptote?

6. Continuity and the Intermediate Value Theorem.

- (a) For each property, sketch the graph of a function f defined on $[0, 4]$ with that property, and name the type of discontinuity at $x = 2$. (i) $\lim_{x \rightarrow 2} f(x) = 3$ but $f(2) = 1$. (ii) $\lim_{x \rightarrow 2^-} f(x) = 1$ and $\lim_{x \rightarrow 2^+} f(x) = 4$. (iii) $\lim_{x \rightarrow 2^+} f(x) = -\infty$.
- (b) Let $f(x) = \begin{cases} x + 2 & x < -1 \\ x^2 & -1 \leq x \leq 1 \\ 3 - x & x > 1 \end{cases}$. Sketch the graph of f . Use the definition of continuity to decide whether f is continuous at $x = -1$ and at $x = 1$. At which x values is f not differentiable?
- (c) Where is $g(x) = \frac{\sqrt{x-3}}{x^2-25}$ continuous?
- (d) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.
- (e) Show that $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

7. The derivative as a limit.

- (a) Let $f(x) = x^3$ and $P = (2, 8)$. Write the slope of the secant line through P and $Q = (x, x^3)$. Take the limit as $x \rightarrow 2$ to find the slope of the tangent line at P , then write the tangent line. Sketch the curve, P , one secant line, and the tangent line.
- (b) Compute $f'(x)$ directly from $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ for $f(x) = \frac{x}{x+2}$.
- (c) Compute $f'(x)$ from the same limit for $f(x) = \sqrt{2x+1}$. Then find the tangent line at $(4, 3)$.
- (d) A ball's height is $h(t) = 200 - 16t^2$ feet after t seconds. Find the slope of the secant line between $t = 1$ and $t = 3$, with units. Then find the instantaneous velocity at $t = 2$ as a limit.
- (e) Show that $f(x) = \sqrt[3]{x}$ is continuous at $x = 0$ but not differentiable there. What does the tangent line at $x = 0$ look like?
- (f) Each limit is $f'(a)$ for some function f and number a . Find f and a , then evaluate:
 $\lim_{h \rightarrow 0} \frac{(2+h)^4 - 16}{h}$ and $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \frac{\pi}{2}}$.
- (g) Sketch the graph of f' for the function f graphed below. Where is f' zero, positive, negative?



8. Differentiation rules. Differentiate, using the power, product, and quotient rules.

- (a) Find $f'(x)$ if $f(x) = 6x^4 - \frac{2}{x^2} + 3\sqrt[3]{x} - 7$
- (b) Find $g'(x)$ if $g(x) = (x^3 + 2x)(4 - x^2)$
- (c) Find y' if $y = \frac{x^2+1}{2x+1}$
- (d) Find the points on the curve $y = x^3 - 3x^2 - 9x + 2$ where the tangent line is parallel to the line $y = 15x$.
- (e) Find the tangent line to $f(x) = x^3 - 4x^2 + 5$ at $x = 3$.
- (f) A particle has position $s(t) = t^3 - 9t^2 + 24t$ meters at time t seconds. Find the velocity $v(t)$ and acceleration $a(t)$, with units. When is the particle at rest?

9. Trig derivatives.

- (a) Starting only from $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$, derive the formula for $\frac{d}{dx} \sec x$. Then find $\frac{d^2}{dx^2} \tan x$.
- (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$.
- (c) Find $f'(\theta)$ if $f(\theta) = \theta^2 \cos \theta - 4 \tan \theta$
- (d) Find y' if $y = \frac{\sec x}{1 + \tan x}$
- (e) Find the tangent line to $y = x \cos x$ at the point $(\pi, -\pi)$.

10. Chain rule.

- (a) Use the table to find $(f \circ g)'(1)$, $(g \circ f)'(1)$, and $(fg)'(2)$.

x	0	1	2	3
$f(x)$	2	3	0	1
$f'(x)$	-1	4	2	5
$g(x)$	1	2	3	0
$g'(x)$	3	-2	1	4

- (b) Find the tangent line to $y = \tan(\sqrt{x})$ at $x = \frac{\pi^2}{16}$.
 (c) Find the tangent line to $y = (x^2 - 3)^5$ at $x = 2$.
 (d) Find $f'(x)$ if $f(x) = \sqrt{x^4 + 3x^2 + 5}$
 (e) Find $g'(x)$ if $g(x) = 3x^2 \sec(x^3 - 1)$
 (f) Find $h'(x)$ if $h(x) = \frac{\cos^2(x)}{(2x+3)^3}$
 (g) Find y' if $y = \sin((3x^2 - 1)^4)$
 (h) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = 2 \sin x + \cos(2x)$ has a horizontal tangent line.

11. Implicit differentiation.

- (a) For the ellipse $x^2 + 4y^2 = 16$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Simplify $\frac{d^2y}{dx^2}$ using the equation of the ellipse.
 (b) Find the tangent line to $\sin(x + y) = 2x - 2y$ at (π, π) .
 (c) Find the tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$.
 (d) For the curve $y^2 = x^3 - 3x + 2$, find $\frac{dy}{dx}$, the tangent line at $(2, 2)$, and all points with a horizontal tangent line.

12. Related rates. Draw a picture, name the quantities that change, and write the equation relating them before differentiating.
- (a) A 13 ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. How fast is the top sliding down when the bottom is 5 ft from the wall?
 - (b) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at $4 \text{ m}^3/\text{min}$. How fast is the water level rising when the water is 4 m deep?
 - (c) The height of a cylinder increases at 3 cm/min while its radius decreases at 0.5 cm/min. How fast is the volume $V = \pi r^2 h$ changing when $r = 2 \text{ cm}$ and $h = 4 \text{ cm}$? Is it increasing or decreasing?
 - (d) A 6 ft tall person walks away from a 15 ft lamppost at 3 ft/s. How fast is the tip of the person's shadow moving?
 - (e) A marble moves along the curve $x^2 + xy + y^2 = 9$ (x, y in feet). At $(3, 0)$, x is decreasing at 2 ft/s. Is y increasing or decreasing, and how fast?
13. Linear approximation and differentials.
- (a) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.
 - (b) Find the linearization of $f(x) = \sin x$ at $a = \frac{\pi}{6}$. Use it to approximate $\sin\left(\frac{\pi}{6} + 0.02\right)$.
 - (c) The radius of a sphere is measured as 10 cm with a possible error of 0.05 cm. Use differentials to estimate the maximum error in the computed volume.

14. Extreme values.

- (a) Find the critical numbers of $f(x) = x^{2/3}(x - 5)$. Which come from $f'(x) = 0$ and which from $f'(x)$ undefined?

Find the absolute extrema of f on the given interval.

- (b) $f(x) = x^4 - 8x^2 + 3$ on $[-1, 3]$
(c) $f(x) = \frac{4x}{x^2+4}$ on $[-4, 0]$
(d) $f(x) = x\sqrt{4-x^2}$ on $[-1, 2]$
(e) $f(x) = 2\cos x + \sin(2x)$ on $[0, \pi]$

15. Mean Value Theorem.

- (a) Find every c guaranteed by the MVT for $f(x) = x^2 - 4x + 1$ on $[1, 5]$.
(b) If $f(2) = -1$ and $f'(x) \leq 3$ for all x , how large can $f(6)$ be?
(c) Show that $f(x) = |x|$ on $[-1, 1]$ has no c with $f'(c) = \frac{f(1)-f(-1)}{1-(-1)}$. Which hypothesis of the MVT fails?
(d) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

16. Shape of a graph. Find the intervals of increase and decrease, the local extrema, the intervals of concavity, and the inflection points.

(a) $f(x) = x^3 - 6x^2 + 9x + 1$

(b) $f(x) = 3x^4 - 4x^3$

(c) $f(x) = x + 2\sin x$ on $[0, 2\pi]$

(d) Use the second derivative test to classify the critical numbers of $f(x) = x^4 - 2x^2$.

(e) Sketch a function with $f' > 0$ and $f'' < 0$ everywhere. Sketch one with $f' < 0$ and $f'' > 0$ everywhere.

(f) Find constants a and b so that $f(x) = x^3 + ax^2 + bx$ has an inflection point at $x = 1$ and a local minimum at $x = 3$. Where is its local maximum?

17. Curve sketching. Find the domain, intercepts, asymptotes, intervals of increase and decrease, local extrema, concavity, and inflection points. Then sketch the graph and label all of these.

(a) $f(x) = \frac{1}{x^2+3}$

(b) $f(x) = \frac{x^2}{x-1}$ (this one has a slant asymptote)

(c) $f(x) = x - 2\sqrt{x}$

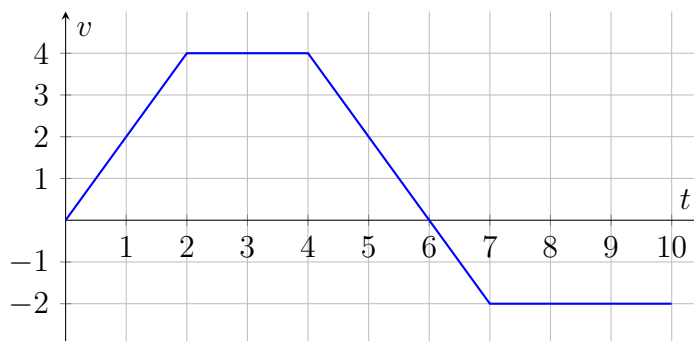
18. Optimization. Draw a picture, introduce variables, write the quantity to optimize as a function of one variable, and verify that your answer is a max or min.
- (a) A closed box has a rectangular base whose length is twice its width, and its volume is 72 ft^3 . Find the dimensions that use the least material.
 - (b) A closed cylindrical can holds $16\pi \text{ cm}^3$. Find the radius and height that minimize its surface area.
 - (c) A rectangle has its base on the x -axis and its two upper corners on the semicircle $y = \sqrt{25 - x^2}$. Find the largest possible area.
 - (d) A rectangular pen is split in half by a divider parallel to one side. The outer fence costs \$4 per meter and the divider costs \$2 per meter. With \$200, what is the largest possible area?
 - (e) Find the points on the parabola $y = x^2$ closest to the point $(0, 2)$.
19. Antiderivatives and initial value problems. Find the most general antiderivative.
- (a) $f(x) = 6x^2 - \frac{4}{x^3} + 2\sqrt{x}$
 - (b) $g(x) = \frac{x^2 - 3x + 2\sqrt{x}}{x}$
 - (c) $h(t) = 3 \cos t - 2 \sec^2 t + \sec t \tan t$
 - (d) If $f'(x) = 3\sqrt{x} - 2$ and $f(4) = 10$, find $f(x)$.
 - (e) A particle has acceleration $a(t) = 6t - 4$, initial velocity $v(0) = 3$, and initial position $s(0) = 1$. Find $s(t)$.
 - (f) Let F be the antiderivative of $f(x) = x^2 - 4$ with $F(0) = 0$. Without finding F , decide where F is increasing, where it has local extrema, and where it is concave up. Then find F and sketch it.

20. Riemann sums.

- (a) Estimate $\int_0^\pi \sin x \, dx$ with the midpoint rule and $n = 3$. Graph $\sin x$ and draw each rectangle.
- (b) Approximate the area under $f(x) = x^2$ on $[0, 4]$ with $n = 4$ rectangles using right endpoints, left endpoints, and midpoints. Which are over estimates, and which are under estimates?
- (c) Write $\int_1^4 (x^2 + 3x) \, dx$ as a limit of Riemann sums using right endpoints. Do not evaluate.
- (d) Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

21. The definite integral as area. Compute each integral with geometry (areas of triangles, rectangles, and circles), not antiderivatives.

- (a) $\int_0^2 (3x + \sqrt{4 - x^2}) \, dx$
- (b) $\int_{-2}^2 (x^3 + \sqrt{4 - x^2}) \, dx$
- (c) $\int_{-3}^3 (|x| - 1) \, dx$
- (d) If $\int_0^5 f(x) \, dx = 7$ and $\int_0^2 f(x) \, dx = 3$, find $\int_2^5 f(x) \, dx$, $\int_5^2 f(x) \, dx$, and $\int_2^5 (3f(x) - 1) \, dx$.
- (e) The velocity $v(t)$ (in m/s) of a particle moving on a line is graphed below, for t in seconds.



- i. What is the displacement of the particle from $t = 0$ to $t = 10$?
- ii. What is the total distance traveled from $t = 0$ to $t = 10$?
- iii. If the particle is at $x = 3$ m at $t = 0$, where is it at $t = 10$?
- iv. What is the acceleration at $t = 5$? When is the particle farthest from where it started?

22. Fundamental Theorem of Calculus, Part 1. Find the derivative.

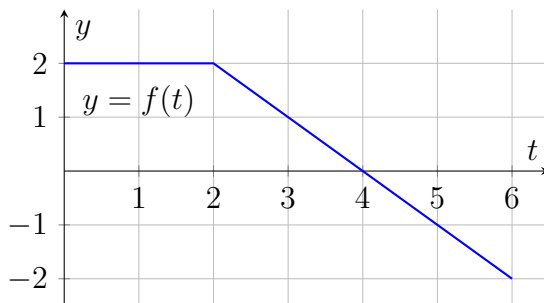
(a) $g(x) = \int_1^x \sqrt{t^3 + 2} \, dt$

(b) $g(x) = \int_{x^2}^4 t\sqrt{1+t^2} \, dt$

(c) $F(x) = \int_0^{\tan x} \sqrt{1+t^2} \, dt$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (simplify)

(d) $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$

(e) Let $g(x) = \int_0^x f(t)dt$, where f is graphed below. Find $g(4)$, $g'(4)$, and $g''(4)$. Where on $[0, 6]$ does g have its maximum?



23. Fundamental Theorem of Calculus, Part 2. Evaluate the integral.

(a) $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2}\right) dx$

(b) $\int_0^{\pi/4} \sec x \tan x \, dx$

(c) $\int_0^\pi (\sin x + 2 \cos x) dx$

(d) What is wrong with the calculation $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

24. Substitution. Clearly declare each substitution. For definite integrals, change the bounds to u -bounds.

(a) $\int \sec^2 x \tan^3 x \, dx$

(b) $\int_0^{\pi/2} \cos x \sqrt{1 + 3 \sin x} \, dx$

(c) $\int \sqrt{x} \sin(x^{3/2}) \, dx$

(d) $\int_0^4 \frac{x}{\sqrt{x^2+9}} dx$

(e) $\int_{-2}^1 x \sqrt{x+3} \, dx$

(f) $\int_{-\pi/2}^{\pi/2} x^2 \sin x \, dx$ (look for symmetry first)

25. Area between curves. Sketch the region and find its area.

(a) $y = x^2 - 2x$ and $y = x + 4$

(b) $y = \sqrt{x}$ and $y = \frac{x}{3}$. Find the area integrating with respect to x , then with respect to y .

(c) $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$

26. Average value.

(a) Find the average value of $f(x) = \frac{1}{x^2}$ on $[1, 3]$.

(b) Find the average value of $f(x) = x^2$ on $[0, 3]$, and find c in $[0, 3]$ with $f(c)$ equal to the average value.

27. Volumes. Sketch the region and a representative slice. Set up an integral for the volume, then evaluate it.
- (a) Set up, but do not evaluate. (i) The region under $y = \sin x$ for $0 \leq x \leq \pi$, revolved around the line $y = -1$. (ii) The region between $y = x^2$ and $y = 1$, revolved around the line $y = 1$.
 - (b) The region between $y = x$ and $y = x^2$, revolved around the x -axis
 - (c) The region under $y = \sqrt{x}$ for $0 \leq x \leq 4$, revolved around the x -axis
 - (d) The region between $y = \sqrt{x}$ and $y = x^2$, revolved around the y -axis, and then around the line $x = -1$
28. Work.
- (a) A spring has natural length 20 cm, and 0.9 J of work stretches it from 20 cm to 26 cm. How much work stretches it from 24 cm to 28 cm? What force holds it at a length of 30 cm?
 - (b) A spring has natural length 10 cm and spring constant k . Find the work W_1 to stretch it from 10 cm to 15 cm and the work W_2 to stretch it from 15 cm to 20 cm, in terms of k . How are W_1 and W_2 related?
 - (c) A 10 m chain with mass 2 kg per meter hangs from the top of a building. How much work does it take to pull the whole chain up to the top? Use $g = 9.8 \text{ m/s}^2$.
 - (d) A force of $F(x) = \frac{x}{(x^2+1)^2}$ pounds acts on an object x feet from the origin. Find the work done moving the object from $x = 0$ to $x = 1$.